MIL-F-20531A (MU)
19 May 1967
SUPERSEDING
MIL-F-20531
7 December 1951
PA-PD-2219 (REV.1)
30 July 1963

M

MILITARY SPECIFICATION

FLARE, AIRCRAFT, PARACHUTE M9A1

1. SCOPE

1.1 This specification covers parts and the loading, assembling, and packing of one type of flare designated as Flare, Aircraft, Parachute, M9A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-31 - Paper; General Specification on Methods of Testing

MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-12951 - Primer, Percussion, M39A1, Loading, Assembling and Packing
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

FSC: 1370

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8848573 - Flare, Aircraft, Parachute, M9A1, Assembly
8850201 - Box, Packing, Ammunition, for Flare, Aircraft, Parachute, M9A1
8850202 - Container, Ammunition, Metal, for Flare, Aircraft, Parachute, M9A1

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET-8848573 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Flare assembly.-The flare assembly shall comply with all requirements specified on Drawing (dwg.) 8848573, and with all requirements specified in applicable specifications.

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3.3 Moisture content of illuminant composition and fuze composition.-The moisture content of the illuminant composition and the fuze composition at the loading station at the time of loading shall not exceed 0.30 percent when tested as specified in 4.4.1.

3.4 Functioning (Proving Ground).-The flare assembly shall function and comply with the following requirements when tested as specified in 4.4.2.

3.4.1 Premature.-The flare assembly shall not:

(1) exhibit a perceptible delay between initiation of the primer and ejection from the projector.
(2) burst in the projector.
(3) eject with insufficient velocity to clear plane, or
(4) eject the parachute and candle assembly in less than one and one half seconds after initiation of the primer.

3.4.2 The candle shall ignite and burn for the required time.

3.4.3 The parachute shall show no evidence of incomplete opening or broken or twisted shroud lines while suspending the burning candle.

3.4.4 The parachute and candle assembly shall eject from the flare case in not more than 3 seconds after primer initiation.

3.4.5 During the minimum (min.) burning time, no burning pieces of the candle assembly that burn 5 seconds or longer shall become detached.

3.5 Leakage.-The sealed metal container shall not leak, as evidenced by emerging bubbles, when tested as specified in 4.4.3.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

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3.7 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Statements of findings on all compositions and powders and certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.
h. A statement that the components and assemblies were dried and assembled as required by the applicable drawing.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit for evaluation a first article sample to an approved laboratory and proving ground designated by the contracting officer. The first article sample shall consist of the following quantities of inert and loaded parts, sub-assemblies, and assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract.

a. Fifteen (15) complete sets of inert parts (i.e. 15 each of every inert component and inert sub-assembly) shall be submitted to a Government approved laboratory designated by the contracting officer. (see 6.1c) for inspection in accordance with all applicable drawings and specifications.

b. Fifteen (15) loaded assemblies designated by the contract, completely packaged and sealed as required by the applicable drawings, shall be submitted for testing to a Government proving ground designated by the contracting officer.

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c. Fifteen (15) illuminant assemblies shall be submitted for candlepower test to Picatinny Arsenal, Dover, New Jersey.

All parts and materials, including packaging and packing components shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials (see 4.1.1e). A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. Unless otherwise directed by the contracting officer, loaded components shall not be submitted for any test until the inert component first article sample has been accepted. All tests shall be performed in accordance with the equipment and methods specified in 4.4.

4.2.2 Inspections to be performed.-Prior to the submission of the loaded assemblies to the proving ground for functioning test and the illuminant assemblies to Picatinny Arsenal for candlepower test, the loaded assemblies shall be inspected by the Government at the contractor's plant for all the requirements of the drawing and specification that are non-destructive and do not require any disassembly. The quantity of loaded components and illuminant assemblies to be subjected to test shall be as follows:

a. Leakage - 15 assemblies in sealed metal containers (see 3.5)
b. Functioning - 15 assemblies (the same assemblies as tested for leakage (see 3.4)
c. Candlepower - 15 illuminant assemblies (see dwg. 8848505).

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4.2.3 Rejection.-If any inert component, loaded assembly or illuminant assembly fails to comply with any or all of the specified requirements the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an item in the sample to comply with any or all requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample, or portion thereof, of items for first article inspection. Until that phase of the first article sample consisting of inert components and loaded items is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Loaded primers from not more than one lot interfix number.
b. Black powder from not more than one lot.
c. Parts of one kind from one manufacturer.
d. Quick match of one lot number from one manufacturer.
e. Composition ingredients from not more than one lot.
f. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

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Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects (and Majors, when so specified) shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in Military Standard MIL-STD-105. Continuous sampling plans in accordance with Standard MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Barrel (see dwg. 8848516 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Diameter of primer hole.....	Gage	01001
102.	Diameter of charge cavity.....	Gage	01002
103.	Length between retainer seat and base.....	Gage	01003
104.	Thickness through base.....	Gage	01004
105.	Outside diameter.....	Gage	01005

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Minor:	AQL 4.00 percent		
201.	Total length	Gage	01006
202.	Diameter of primer hole counterbore	Gage	01007
203.	Depth of primer hole counterbore	Gage	01008
204.	Distance from base to groove	Gage	01009
205.	Width of groove	Gage	01010
206.	Diameter of groove	Gage	01011
207.	Cut or deep scratches on interior	Visual	01012
208.	Evidence of poor workmanship (see 3.6)	Visual	01013

4.3.2.2 Adapter, Fuze (see dwg. 8848568 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Pitch diameter of thread, min	Gage	02001
102.	Major diameter, min	Gage	02002
103.	Thread length, min	Gage	02003
104.	Inside diameter, max	Gage	02004
105.	Total length, min	Gage	02005
Minor:	AQL 2.50 percent		
201.	Diameter of counterbore	Gage	02006
202.	Depth of counterbore	Gage	02007
203.	Diameter of flange	Gage	02008
204.	Thickness of flange	Gage	02009
205.	Length between two flat surfaces	Gage	02010
206.	Crack or deep scratch	Visual	02011
207.	Evidence of poor workmanship (see 3.6)	Visual	02012

4.3.2.3 Housing, fuze (see dwg. 8848523 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Outside diameter.....	Gage	03001
102.	Diameter of flange, max.....	Gage	03002
Minor:	AQL 1.00 percent		
201.	Wall thickness.....	Gage	03004
202.	Evidence of poor workmanship (see 3.6).....	Visual	03005

4.3.2.4 Fuze Assembly (see dwg. 8848524 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Length from the end to flange, max.....	Gage	04001
102.	Housing cracked or with deep scratch.....	Visual	04002
Minor:	AQL 2.50 percent		
201.	Outside diameter of neck.....	Gage	04003
202.	Length to datum diameter.....	Gage	04004
203.	Cutting of notches missing.....	Visual	04005
204.	Grooves on fuze composition missing.....	Visual	04006
205.	Evidence of poor workmanship (see 3.6).....	Visual	04007

4.3.2.5 Cup, Expelling Charge (see dwg. 8848520 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Pitch diameter of thread, max....	Gage	05001
102.	Minor diameter of thread, max....	Gage	05002
103.	Crack or deep scratch on interior.	Visual	05003
Minor:	AQL 2.50 percent		
201.	Diameter of large counterbore....	Gage	05004
202.	Diameter of small counterbore....	Gage	05005
203.	Depth of large counterbore.....	Gage	05006
204.	Depth to bottom of small counterbore.....	Gage	05007
205.	Outside diameter.....	Gage	05008
206.	Overall length.....	Gage	05009
207.	Evidence of poor workmanship (see 3.6).....	Visual	05010

4.3.2.6 Expelling Charge Cup Assembly (see dwg. 8848522 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Charge missing.....	Visual	06001
102.	Crimp of wad and drumhead not full 360 degrees.....	Visual	06002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	06003

4.3.2.7 Head, Flare Case, alternative (see dwg. 8848567 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Diameter of crimping lip, min....	Gage	07001
102.	Wall thickness, min.....	Gage	07002
103.	Length of crimping lip, min.....	Gage	07003
Minor:	AQL 2.50 percent		
201.	Thickness of bottom, min.....	Gage	07004
202.	Diameter of hole.....	Gage	07005
203.	Total length.....	Gage	07006
204.	Length of straight sides.....	Gage	07007
205.	Diameter of chamfered corners, min.....	Gage	07008
206.	Evidence of poor workmanship (see 3.6).....	Visual	07009

4.3.2.8 Case, Flare (see dwg. 8848570 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Inside diameter.....	Gage	08001
102.	Wall thickness at open end, min..	Gage	08002
103.	Wall thickness at rear end, min..	Gage	08003
104.	Diameter of hole, max.....	Gage	08004
105.	Crack, fracture, or dent.....	Visual	08005
Minor:	AQL 2.50 percent		
201.	Total length.....	Gage	08006
202.	Length to shoulder.....	Gage	08007
203.	Outside diameter of rear end at shoulder.....	Gage	08008
204.	Outside diameter at rear end.....	Gage	08009
205.	Thickness at shoulder.....	Gage	08010
206.	Evidence of poor workmanship (see 3.6).....	Visual	08011

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4.3.2.9 Case, Flare (alternative) (see dwg. 8848569 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Inside diameter	Gage	09001
102.	Wall thickness, min	Gage	09002
103.	Diameter of hole, max	Gage	09003
104.	Crack or fracture	Visual	09004
Minor:	AQL 1.00 percent		
201.	Total length	Gage	09005
202.	Evidence of poor workmanship (see 3.6)	Visual	09006

4.3.2.10 Flare Case Assembly or Flare Case Assembly, alternative prior to assembling expelling charge cup assembly (see dwg. 8848571 or 8848572 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Weight of expelling charge	Balance	10001
Minor: None defined			

4.3.2.11 Flare Case Assembly (see dwg. 8848571 or 8848572 covering a detail of dwg. 8848573 and Note 1).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Any operation missing.....	Visual	11001
102.	Any component missing, insecure, or damaged to extent that function may be impaired.....	Visual	11002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	11003

4.3.2.12 Cap, closing (see dwg. 8848515 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Crack or fracture.....	Visual	12001
Minor:	AQL 1.50 percent		
201.	Wall thickness.....	Gage	12002
202.	Length.....	Gage	12003
203.	Evidence of poor workmanship (see 3.6).....	Visual	12004

4.3.2.13 Cap Cover, (see dwg. 8848500 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Diameter of hole.....	Gage	13001
202.	Depth from base to flange.....	Gage	13002
203.	Evidence of poor workmanship (see 3.6).....	Visual	13003

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4.3.2.14 Tube cover (see dwg. 8848501 covering a detail of 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Outside diameter	Gage	14001
202.	Inside diameter	Gage	14002
203.	Evidence of poor workmanship (see 3.6)	Visual	14004

4.3.2.15 Cover Assembly (see dwg. 8848502 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cover cap missing, damaged, or crimp not full 360 degrees	Visual	15001
Minor:	AQL 1.50 percent		
201.	Width of crimped flange, min	Gage	15002
202.	Total length	Gage	15004
203.	Evidence of poor workmanship (see 3.6)	Visual	15003

4.3.2.16 Canopy, Parachute (see dwg. 8848578 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Width of canopy.....	Scale	16001
Minor:	AQL 1.00 percent		
201.	Seaming incorrect.....	Visual	16002
202.	Evidence of poor workmanship (see 3.6).....	Visual	16003

4.3.2.17 Parachute Assembly (see dwg. 8848579 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cloth defective (stain, tear, cut, hole, etc.).....	Visual	17001
102.	Pinks missing.....	Visual	17002
Minor:	AQL 2.50 percent		
201.	Length of shroud line.....	Scale	17003
202.	Distance from tying twine to center line of canopy.....	Scale	17004
203.	Length of reinforcing tape.....	Scale	17005
204.	Center shroud line not knotted at end or incorrectly stitched...	Visual	17006
205.	Shroud lines not knotted at ends or incorrectly tied to eyelets...	Visual	17007
206.	Tying cord does not bind all shroud lines securely together...	Visual	17008
207.	Evidence of poor workmanship (see 3.6).....	Visual	17009

4.3.2.18 Spacer, Flare Case (see dwg. 8848507 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Outside diameter, min.....	Gage	18001
102.	Diameter of hole.....	Gage	18002
Minor:	AQL 1.50 percent		
201.	Diameter of hole counterbore.....	Gage	18003
202.	Depth of hole counterbore.....	Gage	18004
203.	Total length.....	Gage	18005
204.	Evidence of poor workmanship (see 3.6).....	Visual	18006

4.3.2.19 Spacer Assembly (see dwg. 8848508 and covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Quickmatch missing.....	Visual	19001
102.	End of quickmatch above flush....	Visual	19002
103.	Spacer cracked or split.....	Visual	19003
Minor:	AQL 1.00 percent		
201.	Coil end incorrect.....	Visual	19004
202.	Evidence of poor workmanship (see 3.6).....	Visual	19005

4.3.2.20 End, Candle Case (see dwg. 8848497 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Hole missing.....	Visual	20003

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Minor:	AQL 1.50 percent		
201.	Thickness of side wall.....	Gage	20004
202.	Length.....	Gage	20005
203.	Evidence of poor workmanship (see 3.6).....	Visual	20006

4.3.2.21 Case, Candle (see dwg. 8848498 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Outside diameter.....	Gage	21001
102.	Wall thickness.....	Gage	21002
103.	Cuts, gauges, blisters, or other imperfections.....	Visual	21003
104.	Exterior coating missing.....	Visual	21004
Minor:	AQL 1.00 percent		
201.	Total length.....	Gage	21005
202.	Evidence of poor workmanship (see 3.6).....	Visual	21006

4.3.2.22 Candle Case Assembly (see dwg. 8848499 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Total length, max.....	Gage	22001
102.	End missing or incompletely sealed.....	Visual	22002
103.	Assembly damaged (cuts, puncture, soft spots, wrinkles or layers separating.....	Visual	22003

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	22004

4.3.2.23 Candle Assembly, prior to assembling loaded candle case assembly (see dwg. 8848505 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Parachute wire knot incorrect....	Visual	23001
Minor:	None defined		

4.3.2.24 Candle Assembly, prior to assembling cover assembly (see dwg. 8848505 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Charge or clay header missing....	Visual	24001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	24002

4.3.2.25 Candle assembly (see dwg. 8848505 covering a detail of dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Candle case assembly loose in cover assembly.....	Manual	25001
102.	Parachute wire loop wrap improper, or tape missing or insecure.....	Visual-Manual	25002
Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	25003
202.	Assembly dented.....	Visual	25004

4.3.2.26 Flare assembly, prior to assembling flare case assembly to barrel (see dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Propelling charge missing.....	Visual	26001
Major:	AQL 0.65 percent		
101.	Diameter of barrel at mouth, max.	Gage	26002
102.	Primer missing or above flush....	Visual	26003
103.	Spacer missing or inverted.....	Visual	26004
104.	Quickmatch missing or damaged to extentthat function may be impaired.....	Visual	26005
Minor:	AQL 1.50 percent		
201.	Wad or propelling charge missing or insecure.....	Visual	26006
202.	Evidence of poor workmanship (see 3.6).....	Visual	26007
203.	Tallow coating missing or excessive.....	Visual-Manual	26008

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4.3.2.27 Flare Assembly, prior to assembling candle assembly (see dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Set-back wad missing.....	Visual	27001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	27002

4.3.2.28 Flare Assembly, prior to inserting parachute assembly (see dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Gas check, gas check disc, or thrust member missing.....	Visual	28001
102.	Assembly of parachute to candle assembly incorrect or insecure.....	Visual-Manual	28002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	28003

4.3.2.29 Flare Assembly, prior to assembling closing cap (see dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Parachute disc or cap reinforcing disc missing.....	Visual	29001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	29002

4.3.2.30 Flare Assembly (see dwg. 8848573).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Length, max.....	Gage	30001
102.	Joint not sealed.....	Visual	30002
103.	Assembly damaged to extent that function may be impaired.....	Visual	30003
104.	Barrel loose on flare case assembly.....	Manual	30004
Minor:	AQL 1.50 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	30005
202.	Marking or stamping incorrect or unidentifiable.....	Visual	30006
203.	Closing cap not dry.....	Visual-Manual	30007

4.3.2.31 Container, Metal, Sealed (see dwg. 8850202).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Puncture or deep scratches.....	Visual	31001
102.	Container dented or deformed.....	Visual	31002
103.	Key missing.....	Visual	31003
Minor:	AQL 1.50 percent		
201.	Contents move when container is shaken.....	Manual	31004
202.	Bare spot on exterior other than slight scratches.....	Visual	31005
203.	Marking misleading or unidentifiable.....	Visual	31006
204.	Evidence of poor workmanship (see 3.6).....	Visual	31007

4.3.2.32 Box, packing, sealed (see dwg. 8850201).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Hardware missing, broken or loose.....	Visual-Manual	32001
102.	Strapping missing, broken or loose.....	Visual-Manual	32002
103.	Contents escaped.....	Visual	32003
104.	Board broken or split.....	Visual	32004
105.	DOD symbol misleading or unidentifiable.....	Visual	32005
Minor:	AQL 2.50 percent		
201.	Hardware improperly engaged.....	Visual-Manual	32006
202.	Handle missing or insecure.....	Visual-Manual	32007
203.	Contents loose (when box is tilted).....	Manual	32008

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204.	Strapping improperly assembled...	Visual	32009
205.	Marking misleading or unidentifiable.....	Visual	32010
206.	Car seal, missing, unsealed or improperly positioned.....	Visual	32011
207.	Evidence of poor workmanship (see 3.6).....	Visual	32012

4.3.3 Production testing.-Unless otherwise specified herein, the following tests shall be performed using Standard MIL-STD-105 for sampling plan requirements.

4.3.3.1 Moisture content of illuminant composition, fuze composition, black powder, gas check, set back wad, parachute disc, and wad (see Table I).-The contractor shall provide adequate controls to insure that all the powders and components specified in Table I comply with the moisture content requirements. For verification, and just prior to start of loading the contractor shall select one sample of sufficient size to perform each test from that quantity or sub-lot of each powder or components used in eight hours loading production of flare assemblies. A composite sample shall not be used. If the moisture content of the sample exceeds the requirement and loading has not begun, that sub-lot of powder or component represented by the sample shall be rejected. If powders or components from a sub-lot with excessive moisture have been loaded, all flares assemblies containing the non-conforming powder or components shall be rejected. The tests shall be performed as specified in 4.4.1.

TABLE I

Item	Classification	Code No.
Illuminant composition (see 3.3)	Major	33001
Fuze composition (see 3.3)	Major	33002
Black powder (see dwg. 8848573)	Major	33003
Gas check (see dwg. 8848573)	Major	33004
Set back wad (see dwg. 8848573)	Major	33005
Parachute disc (see dwg. 8848573)	Major	33006
Wad (see dwg. 8848573)	Major	33007

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4.3.3.2 Functioning (see Table II).

TABLE II

Defects	Classification	Code No.
Premature (see 3.4.1)	Critical	30008
Candle ignition and burning time (see 3.4.2)	Major	30009
Parachute suspension (see 3.4.3)	Major	30010
Parachute and candle ejection (see 3.4.4.)	Major	30011
Candle composition break-up (see 3.4.5)	Major	30012

4.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 80 samples shall be selected from each lot for test. If a critical defect occurs, or if 3 or more flare assemblies exhibit a major defect, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.2.2 After three consecutive lots have met the criteria of 4.3.3.2.1 a single sampling plan in accordance with code letter G and an AQL of 1.50 percent shall be used. The lot shall be rejected if during the test a critical defect occurs, or the number of assemblies that exhibit a major defect equals or exceeds the sampling plan rejection number.

4.3.3.3 Leakage (see 3.5) Major defect - Code No. 31008.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.3. Any container that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.4 Candlepower (see dwg. 8848505)-Major defect - Code No. 25005.-The sampling plan for this test shall be in accordance with Code letter G, and an AQL of 2.50 percent. The test shall be performed as specified in 4.4.4.

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4.3.3.5 Deterioration check test, Major defect - Code No. 34001.-If the total time between the original acceptance of the primer lot and the assembly of that lot into the flare assemblies exceeds 2 years, or if the primers in the lot have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be check tested as specified in the applicable primer loading specification immediately before the primer lot is assembled into the flare assemblies. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the flare assemblies. The test shall be conducted at Government expense (see 6.5) without cost to the contractor who loaded the primer or the contractor assembling the primers into the flare assemblies, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lot was under the responsibility of either contractor.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-8848573 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with

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contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Illuminant composition.-The two gram sample selected shall be placed in a tared, glass-stoppered weighing dish. The dish and its contents shall then be weighed, dried in an oven at a temperature of 100 degrees Centigrade (C.) plus or minus 2 degrees Centigrade for one hour, cooled in a dessicator and weighed. The loss in weight shall be calculated to percentage of moisture in the composition using the following formula to determine compliance with 3.3.

Percentage of moisture = (A-B) times 100
divided by A

where:
A = weight of dish and contents before drying
B = weight of dish and contents after drying

4.4.1.2 Fuze composition.-The moisture content of the fuze composition shall be determined in accordance with Method 101.2 of Standard MIL-STD-1234.

4.4.1.3 Black powder.-The moisture content of the black powder shall be determined in accordance with Specification MIL-P-223 using a sample weight of approximately 0.5 grams and heating 30 minutes at 70 degrees C.

4.4.1.4 Felt and paper products.-The moisture content of the felt and paper products shall be determined in accordance with Method 240 of Specification UU-P-31.

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4.4.2 Functioning.-The flare assembly shall be fired from an M2 pyrotechnic pistol, or an approved equal, from an airplane flying at an indicated air speed of approximately 200 miles per hour and at an altitude of 1000 feet. Observation shall be made to determine compliance with 3.4.

4.4.3 Leakage.-The sealed container shall be placed in a tank of water and submerged to a depth of 2 inches. The air above the water shall be evacuated to a pressure of 5 to 7 pounds per square inch absolute. After the air inside the tank has been evacuated to the required pressure, the sealed container shall be observed for 15 seconds for any bubbles that would indicate a leak. The test equipment shall be in accordance with 4.3.4.

4.4.4 Candlepower of candle assembly.-The candle assembly shall be removed from the flare assembly, mounted vertically with priming charge end up and ignited. Determination of candlepower shall be made in accordance with the procedure and equipment specified in 4.3.4. The burning time shall be taken for informational purpose only.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The flare assemblies shall be preserved and packaged in accordance with dwg. 8850202.

5.2 Packing

5.2.1 Level A.-The flare assemblies shall be packed in accordance with dwg. 8850201.

5.3 Marking.-Marking of the containers and bases shall be in accordance with dwgs. 8850202 and 8850201.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Submission of first article quantity.-Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Equipment Lists.-Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.5 and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

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MU - Army design, mandatory for use.
CD - Contractor design and/or commercial equipment on contractor format. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection

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equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit two copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Submission of test data.-Instructions as to the location for the evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN. SMUPA-NI.

6.5 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number 1370-A239

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